GENERATIVE CONTACT MECHANICS: A GEOMETRIC FRAMEWORK FOR DISSIPATIVE SYSTEMS WITH STRUCTURED LIMIT CYCLES

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Dedicated to my children Vic (R.I.P.), Giulia, Alice (R.I.P.), Sarah (R.I.P.), and David.

Once tiny, always strong.

Abstract. We introduce a geometric framework for dissipative dynamical systems whose limit cycles carry structured phase, multi-scale coherence, and stochastic stability. The central object is the *dm3 system*—a smooth Riemannian manifold equipped with a hyperbolic limit cycle, a Lyapunov function, and a stochastic extension—formalized through eight axioms. On this foundation we construct a complete operator algebra (the g-, L-, R-, and U-operators) governing expansion, coherence, resonance, and unification, and we embed the entire structure in contact geometry.

Four main results are established. **Theorem A**: any dm3 system admits a unique exponentially stable hyperbolic limit cycle with a topological invariant (the winding integral $\oint_\Gamma \lambda$) and stochastic stability below an embodiment threshold τ. **Theorem B**: the category **dm3** is closed under a unification operator $\mathcal{U}$, with the quantitative prediction $\tau_{12} \leq \min(\tau_1, \tau_2)$: unification weakens noise tolerance. **Theorem C**: every dm3 system near its limit cycle is locally contact-diffeomorphic to a universal three-parameter normal form $(\mu_{\max}, \omega, \beta)$. **Theorem D**: the full operator algebra is C^1-structurally stable with explicit stability radius $\varepsilon_0 = \frac{1}{2}\,|\mu_{\max}| \cdot \sup_\Gamma \|\mathrm{Hess}\, V\|^{-1}$. Complete verification on an explicit toy model appears in the companion paper [14].

Contents

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1. Introduction

1.1. **The problem: dissipative systems with structured limit cycles.** Dissipative dynamical systems with hyperbolic limit cycles appear throughout the natural sciences—in biology as oscillatory cellular and neural processes, in physics as driven nonequilibrium systems, in engineering as controlled oscillators, and in mathematics as paradigmatic examples of nonlinear dynamics. The classical theory provides excellent tools for analyzing a single limit cycle in isolation: Floquet theory characterizes transverse stability, Lyapunov theory gives quantitative convergence rates, and stochastic extensions handle noise.

What is less well understood is how limit cycles interact, unify, and organize into multi-scale coherent structures while remaining robust under perturbation. This paper develops a geometric framework—generative contact mechanics—for precisely this class of behavior. The central insight is that the correct geometric home for dissipative systems with structured limit cycles is not symplectic geometry (which forbids attractors) but contact geometry, where dissipation is encoded in the geometry itself.

1.2. **The contact geometric approach.** A hyperbolic limit cycle is an attractor: nearby trajectories converge to it exponentially. This is incompatible with Hamiltonian mechanics, where Liouville's theorem guarantees that the phase-space volume is preserved and no attractors can exist on compact symplectic manifolds. Standard Lagrangian mechanics is similarly insufficient: the Euler–Lagrange equations are time-reversible and do not admit attractors without additional dissipation terms added by hand.

Contact geometry provides a natural resolution. A contact manifold (M^{2n+1}, α) carries a Reeb vector field and a contact Hamiltonian formalism in which dissipation is built into the geometric structure: the contact Hamiltonian H evolves along trajectories rather than remaining constant, and the action variable $z \in \mathbb{R}$ encodes accumulated dissipation. This makes contact geometry the natural setting for dissipative systems with limit cycle attractors, stochastic stability thresholds, and variational structure [2, 3].

The present paper formalizes this setting through the dm3 system and operator algebra, and shows that the resulting framework is: (a) geometrically natural (contact normal form, Theorem C), (b) categorically coherent (closure of **dm3** under unification, Theorem B), and (c) physically robust (C^1-structural stability, Theorem D).

1.3. **Main results.** The four main theorems are stated informally here and proved in later sections. Precise statements appear at the end of each relevant section.

Theorem 1.1 (A: dm3 existence and stability). *Let $\mathfrak{D} = (S, g, Y, \Gamma, V, \sigma)$ satisfy Axioms 1–8. Then:*

(i) *Γ is a hyperbolic limit cycle of Y with minimal period $T^* > 0$.*
(ii) *Γ is exponentially orbitally stable with rate $c = |\mu_{\max}|$.*
(iii) *For noise amplitude below the embodiment threshold τ, $\mathbb{E}[V(X_t)] \to 0$.*
(iv) *The winding integral $\oint_\Gamma \lambda$ is a topological invariant preserved under all admissible deformations.* (Proved in Proposition 6.3.)

Theorem 1.2 (B: closure under unification). *Let $\mathfrak{D}_1, \mathfrak{D}_2 \in$ **dm3** be in $k{:}m$ resonance with admissible span. Then:*

(i) *The unification operator $\mathcal{U} = U_3 \circ U_2 \circ U_1 \circ R_3 \circ R_2 \circ R_1$ is well-defined and $\mathcal{U}(\mathfrak{D}_1, \mathfrak{D}_2) \in$ **dm3**.*
(ii) *The canonical invariants satisfy $T^*_{12} = kT^*_1 / \gcd(k, m)$, $\mu_{\max,12} = \max(\mu_{\max,i})$, $\tau_{12} \leq \min(\tau_1, \tau_2)$.*
(iii) ***Unification weakens noise tolerance:*** $\tau_{12} \leq \min(\tau_1, \tau_2)$.

Theorem 1.3 (C: contact normal form)**.** *Let $\mathfrak{D}$ satisfy Axioms 1–8. In a tubular neighborhood of Γ in (M,α), there exist contact coordinates (ρ,θ,z) with $\Gamma = \{\rho = 0\}$ such that*

$$\begin{aligned}
\dot\rho &= \mu_{\max}(1 - e^{-\beta z})\rho + O(\rho^2),\\
\dot\theta &= \omega + O(\rho),\\
\dot z &= \omega - |\mu_{\max}|\,\rho^2 e^{-\beta z} + O(\rho^3).
\end{aligned}$$

Every dm3 system is locally contact-diffeomorphic to this normal form. The parameters $(\mu_{\max},\omega,\beta)$ are the canonical invariant triple.

Theorem 1.4 (D: structural stability)**.** *Let $\mathfrak{D}$ satisfy Axioms 1–8 with $\mu_{\max} < 0$. Define*

$$\varepsilon_0 \coloneqq \frac{|\mu_{\max}|}{2\big(1 + \sup_\Gamma \|\mathrm{Hess}\, V\|_\infty\big)}.$$

For any C^1 perturbation F with $\|F\|_{C^1} < \varepsilon_0$, and any finite admissible operator sequence within parameter bounds, the output $\mathfrak{D}'$ satisfies Axioms 1–8 with: a unique hyperbolic limit cycle Γ' near Γ; the same winding number; canonical invariants varying by $O(\varepsilon_0)$; and boundaries B_i' varying C^1-smoothly.

1.4. **Relation to existing work.** *Contact Hamiltonian mechanics.* Bravetti [2] and de León–Lainz [3] developed the modern formalism of contact Hamiltonian mechanics, showing that contact geometry provides a variational framework for dissipative systems. Gaset et al. [4] extended this to Lagrangian contact mechanics. The present paper differs from these works in three respects: (a) we focus on limit cycle attractors rather than fixed-point attractors; (b) we construct a categorical operator algebra acting on contact systems; and (c) we prove structural stability of the operator algebra with an explicit stability radius.

Metriplectic systems. Morrison [11] introduced the metriplectic framework for systems combining Hamiltonian and dissipative dynamics. The dm3 contact formulation is related but distinct: metriplectic structure separates the Poisson and metric brackets, while our contact structure encodes dissipation in the contact form itself through the action variable z.

Normally hyperbolic invariant manifolds. Hirsch, Pugh, and Shub [8] established the persistence and C^r-regularity of normally hyperbolic invariant manifolds under perturbation. Theorem D uses this theory as a foundational input. The novelty is combining NHIM persistence with the operator algebra and establishing the quantitative stability radius ε_0.

Arnold tongues and coupled oscillators. The resonance structure of the R-operators (period resonance, Arnold tongues, mode locking) is classical [1, **?**]. The contribution here is embedding these phenomena in a categorical structure: resonance is the precondition for a well-defined pushout in **dm3**. The prediction $\tau_{12} \le \min(\tau_i)$ is new.

1.5. **Organization.** Section 2 fixes notation and background. Section 3 defines dm3 systems and proves Theorem A. Section 4 constructs the operator algebra and proves Theorem B. Section 5 establishes boundary operators and the operator grammar. Section 6 develops the contact physics and proves Theorem C. Section 7 proves Theorem D. Section 8 discusses open problems and connections. Appendices give the categorical and lemma details.

2. Background

2.1. Riemannian manifolds and flows. Throughout, (S, g) denotes a smooth, connected, complete Riemannian manifold of dimension $n \geq 2$. We write T_xS for the tangent space at x, $\mathfrak{X}(S)$ for smooth vector fields, and $\Omega^k(S)$ for smooth k-forms. For $X \in \mathfrak{X}(S)$ complete, the flow $\phi^t \colon S \to S$ satisfies $\phi^0 = \mathrm{id}$ and $\frac{d}{dt}\phi^t(x) = X(\phi^t(x))$.

2.2. Hyperbolic limit cycles and Floquet theory.

Definition 2.1. A *limit cycle* of $Y \in \mathfrak{X}(S)$ is a compact connected invariant submanifold $\Gamma \cong S^1$ with minimal period $T^* > 0$. It is *hyperbolic* if all Floquet multipliers except the trivial multiplier $\mu_0 = 1$ satisfy $|\mu_j| < 1$ for $j = 1, \ldots, n-1$. The *maximal transverse Lyapunov exponent* is $\mu_{\max} \coloneqq \max_j \frac{1}{T^*} \log |\mu_j| < 0$.

The stable manifold theorem [8] applied to Γ gives: if Γ is hyperbolic, then $\partial\mathcal{B}(\Gamma) = \mathcal{W}^s(\Lambda)$ for some invariant set $\Lambda \subset S \setminus \mathcal{B}(\Gamma)$.

2.3. Lyapunov functions.

Definition 2.2. A smooth function $V \colon S \to \mathbb{R}_{\geq 0}$ is a *Lyapunov function* for Γ if $V(x) = 0 \iff x \in \Gamma$ and $\dot{V}(x) \coloneqq g(\nabla V(x), Y(x)) < 0$ for $x \notin \Gamma$. The standard choice is $V = d_g(\cdot, \Gamma)^2$.

2.4. Stochastic differential equations. For the Itô SDE $dX_t = Y(X_t)\,dt + \sigma(X_t)\,dW_t$ on S, the generator is

$$\mathcal{L}f(x) = g(\nabla f(x), Y(x)) + \tfrac{1}{2}\mathrm{tr}\big(\sigma(x)^\top \,\mathrm{Hess}_f(x)\,\sigma(x)\big).$$

Standard stochastic Lyapunov theory [7] gives: if $\mathcal{L}V \leq -cV + \kappa\,\|\sigma\|^2$ for $c, \kappa > 0$, then $\mathbb{E}[V(X_t)] \to 0$ for noise amplitude below $\tau = \sqrt{c/\kappa}$.

2.5. Contact manifolds.

Definition 2.3. A *contact manifold* (M^{2n+1}, α) is a smooth manifold with a 1-form α satisfying $\alpha \wedge (d\alpha)^n \neq 0$ everywhere. The *Reeb vector field* R_α is the unique vector field with $\iota_{R_\alpha} d\alpha = 0$ and $\alpha(R_\alpha) = 1$. For a smooth $H \colon M \to \mathbb{R}$, the *contact Hamiltonian vector field* X_H is the unique vector field satisfying $\iota_{X_H} d\alpha = dH - (R_\alpha H)\alpha$ and $\alpha(X_H) = -H$.

See Geiges [5] and Bravetti [2] for background.

3. The dm3 operator

3.1. Definition.

Definition 3.1 (dm3 system)**.** A *dm3 system* is a tuple $\mathfrak{D} = (S, g, Y, \Gamma, V, \sigma)$ where (S, g) is a smooth complete Riemannian manifold, $Y \in \mathfrak{X}(S)$ is a smooth vector field with complete flow ϕ^t, $\Gamma \subset S$ is a hyperbolic limit cycle, $V \colon S \to \mathbb{R}_{\geq 0}$ is a smooth Lyapunov function for Γ, and $\sigma \colon S \to \mathbb{R}^{n \times m}$ is a smooth noise coefficient. The *dm3 operator* is $\mathcal{A}_{\mathrm{dm3}} \coloneqq \phi^T$ for $T = T^*/4$.

3.2. **The eight axioms.**

Axiom 1 (Orbit identity)**.** There exist four distinct points $p_I, p_{VI}, p_{III}, p_{VII} \in \Gamma$ traversed cyclically: $\phi^{T^*/4}(p_I) = p_{VI}$, $\phi^{T^*/4}(p_{VI}) = p_{III}$, $\phi^{T^*/4}(p_{III}) = p_{VII}$, $\phi^{T^*/4}(p_{VII}) = p_I$.

Axiom 2 (Generative closure)**.** Γ is compact, connected, and invariant: $\phi^t(\Gamma) = \Gamma$ for all $t \in \mathbb{R}$.

Axiom 3 (Structural primacy)**.** There exists a neighborhood $\mathcal{N}(\Gamma)$ such that $\dot{V}(x) = g(\nabla V(x), Y(x)) < 0$ for all $x \in \mathcal{N}(\Gamma) \setminus \Gamma$.

Axiom 4 (Perturbation response)**.** There exist $c > 0$ and $\mathcal{N}(\Gamma)$ such that $\dot{V}(x) \leq -c\,V(x)$ for all $x \in \mathcal{N}(\Gamma)$.

Axiom 5 (Noise-as-fuel)**.** The SDE generator satisfies $\mathcal{L}V(x) \leq -c\,V(x) + \kappa\,\|\sigma(x)\|^2$ for constants $c, \kappa > 0$.

Axiom 6 (Non-collapse)**.** $Y(x) \neq 0$ for all $x \in \Gamma$, and ϕ^t has minimal period $T^* > 0$ on Γ.

Axiom 7 (Non-coercion)**.** Y is C^∞ on S. No discontinuous projection, clamping, or hard boundary enforcement appears in the definition of Y.

Axiom 8 (Hyperbolicity)**.** All Floquet multipliers of Γ except the trivial one satisfy $|\mu_j| < 1$ for $j = 1, \ldots, n-1$.

Remark 3.2. These eight axioms are logically independent. The original ten axioms are reduced here by: merging the stochastic pair into Axiom 5; and moving invariance under the g- and L-operators and multi-scale compatibility to their respective sections as theorems.

3.3. **Orbital stability.**

Proposition 3.3 (Exponential orbital stability)**.** *Let $\mathfrak{D}$ satisfy Axioms 1–8. Then there exist $c > 0$ and $\mathcal{N}(\Gamma)$ such that $\dot{V}(x) \leq -c\,V(x)$ for all $x \in \mathcal{N}(\Gamma)$.*

Proof. Axiom 8 gives $\mu_{\max} < 0$. By Floquet theory (see [6, 13]), the linearized transverse flow contracts with rate $|\mu_{\max}|$. With $V = d_g(\cdot, \Gamma)^2$, we get $\dot{V} \leq 2\mu_{\max} V + O(V^{3/2})$ in a tubular neighborhood. Taking $\mathcal{N}(\Gamma)$ small enough absorbs the higher-order term, giving $\dot{V} \leq -c\,V$ with $c = |\mu_{\max}|$. □

3.4. **Embodiment threshold and stochastic stability.**

Proposition 3.4 (Stochastic Lyapunov condition)**.** *If Axiom 5 holds with constants c, κ, then for noise amplitude $\|\sigma\| < \tau := \sqrt{c/\kappa}$:*

$$\mathbb{E}[V(X_t)] \leq e^{-ct} V(X_0) + \frac{\kappa}{c} \|\sigma\|^2 .$$

Proof. Standard; see Has'miňskiĭ [7], Chapter 4. □

Definition 3.5 (Embodiment threshold)**.** The *embodiment threshold* is $\tau := \sup\{\sigma_0 : \mathcal{L}V(x) < 0 \text{ whenever } \|\sigma(x)\| < \sigma_0\} = \sqrt{c/\kappa}$. Below τ: noise reinforces the orbit. Above τ: noise disrupts it.

Definition 3.6 (Canonical invariant triple)**.** $\iota(\mathfrak{D}) := (T^*, \mu_{\max}, \tau) \in \mathbb{R}_{>0} \times \mathbb{R}_{<0} \times \mathbb{R}_{>0}$.

3.5. Proof of Theorem A.

Proof of Theorem 1.1. Claims (i)–(ii): Axioms 1, 2, 6, 8 give a compact connected hyperbolic limit cycle with $T^* > 0$. Proposition 3.3 gives exponential stability with $c = |\mu_{\max}|$. Claim (iii): Proposition 3.4 with $\tau = \sqrt{c/\kappa}$. Claim (iv): Proposition 6.3 in Section 6. $\square$

4. The operator algebra

Throughout, $\mathfrak{D} = (S, g, Y, \Gamma, V, \sigma)$ is a fixed dm3 system satisfying Axioms 1–8.

4.1. g-series operators: expansion.

4.1.1. g_1: *local basin expansion.*

Definition 4.1 (g_1)**.** Let $\rho\colon S \to [0,1]$ be a smooth cutoff with $\rho = 0$ on $\mathcal{N}_\delta(\Gamma)$ and $\rho = 1$ outside $\mathcal{N}_{2\delta}(\Gamma)$, and let $h\colon \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be smooth with $\operatorname{supp} h \subset (\delta, R)$ for $R > 2\delta$. Define

$$Z(x) \coloneqq -\rho(x)\, h(V(x))\, \nabla V(x), \quad g_1 \coloneqq \exp(Z) = \psi^1,$$

where ψ^t is the flow of Z.

Proposition 4.2 (Γ-invariance of g_1)**.** $g_1(\Gamma) = \Gamma$.

Proof. For $x \in \Gamma$, $V(x) = 0$, so $\rho(x) = 0$ and $Z(x) = 0$. $\square$

Proposition 4.3 (Basin expansion)**.** *The pushforward $Y^{(1)} \coloneqq (g_1)_* Y$ has Lyapunov function $V^{(1)} = V \circ g_1^{-1}$ satisfying $\dot{V}^{(1)} \leq -c^{(1)} V^{(1)}$ on $\mathcal{N}^{(1)}(\Gamma) \supset \mathcal{N}(\Gamma)$ with $c^{(1)} \geq c$. The expanded system $\mathfrak{D}^{(1)} = (S, g, Y^{(1)}, \Gamma, V^{(1)}, \sigma)$ is a dm3 system with $\tau^{(1)} \geq \tau$.*

Conjecture 4.4 (Global basin expansion)**.** *For appropriate h and δ, the basin of attraction of Γ under $Y^{(1)}$ is strictly larger than under Y.*

4.1.2. g_2: *recursive multi-scale expansion.*

Definition 4.5 (Phase function and second-scale space)**.** The *phase function* $\varphi\colon \mathcal{N}(\Gamma) \to \mathbb{R}/\mathbb{Z}$ assigns arc-length normalized phase, with $\varphi(p_I) = 0$, $\varphi(p_{VI}) = 1/4$, $\varphi(p_{III}) = 1/2$, $\varphi(p_{VII}) = 3/4$. Set $S^{(2)} = \mathbb{R}/\mathbb{Z}$ with projection $\Psi(x) = \varphi(x)$.

Definition 4.6 (g_2 and semi-conjugacy)**.** Let $\omega_2 > 0$, $f\colon S^{(2)} \to \mathbb{R}$ smooth. Define $Y^{(2)}(\Phi) = \omega_2 + f(\Phi)$ with flow $\psi^{(2),t}$, and $g_2 = \psi^{(2),1/\omega_2}$. The orbit Γ is *recursively invariant* if $\exists\, \lambda > 0$ such that

$$\Psi(\phi^t(x)) = \psi^{(2),t/\lambda}(\Psi(x)) \quad \forall\, x \in \Gamma.$$

Setting $\omega_2 = 1/T^*$ and $f \equiv 0$ achieves this with $\lambda = 1$.

Conjecture 4.7 (Invariant torus stability)**.** *If the semi-conjugacy holds and $Y^{(2)}$ has a hyperbolic limit cycle, the product system on $S \times S^{(2)}$ admits a stable invariant torus (KAM-stable in the non-resonant case; mode-locked in the resonant case). Proved in the toy model in* [14].

4.1.3. *g_3: cross-domain expansion and the category* **dm3**.

Definition 4.8 (Category **dm3**). The *category* **dm3** has objects: dm3 systems satisfying Axioms 1–8; morphisms: smooth maps $f: S_1 \to S_2$ satisfying:

(M1) $f(\Gamma_1) = \Gamma_2$,
(M2) $f \circ \phi_1^t = \phi_2^{\lambda t} \circ f$ for some $\lambda > 0$,
(M3) $\alpha\, V_1(x) \le V_2(f(x)) \le \beta\, V_1(x)$ for $0 < \alpha \le \beta < \infty$,
(M4) $\|\sigma_2(f(x))\| \le C\, \|\sigma_1(x)\|$ and $\tau_2 \ge \alpha\tau_1/C$;

composition: function composition; identity: id_S.

Proposition 4.9 (Morphisms preserve dm3 structure). *If $f: \mathfrak{D}_1 \to \mathfrak{D}_2$ is a dm3 morphism, then $\mathfrak{D}_2 \in$* **dm3**.

Proof. Axiom 1: (M1) and (M2) map phase points to phase points. Axiom 2: $f(\Gamma_1) = \Gamma_2$ is invariant by (M2). Axioms 3–4: (M3) gives a Lyapunov function for Γ_2 with rate $c' \ge \alpha c/\beta$. Axiom 5: (M4). Axioms 6–8: smoothness of f and (M1)–(M2) with C^1-diffeomorphism property near Γ_1. □

Corollary 4.10 (Falsifiability of cross-domain claims). *The claim that the dm3 framework applies to a domain D with state space S_D is equivalent to the existence of a morphism $f: \mathfrak{D}_{\text{base}} \to \mathfrak{D}_D$. This is a falsifiable condition.*

4.2. **L-operators: coherence and anti-collapse.**

4.2.1. *L_1: local coherence.*

Definition 4.11 (L_1). With two orbits Γ_1, Γ_2, inter-orbit distance $D_{12}(x) = d_g(x, \Gamma_1) + d_g(x, \Gamma_2)$, and smooth cutoff χ supported where $\min\{d_g(x, \Gamma_i)\} < \varepsilon$, define $Z(x) = -\chi(x)\nabla D_{12}(x)$ and $L_1 = \exp(Z)$.

Proposition 4.12 (L_1 orbit invariance). *$L_1(\Gamma_i) = \Gamma_i$ for $i = 1, 2$. There exists $\varepsilon' > \varepsilon$ such that $d_g(L_1(x), \Gamma_i) \ge d_g(x, \Gamma_i)$ for $d_g(x, \Gamma_i) \in [\varepsilon, \varepsilon']$.*

Proof. $\chi = 0$ on each Γ_i, so $Z = 0$ there. The gradient push increases separation in the buffer zone by compactness. □

Conjecture 4.13 (L_1 hyperbolicity). *For sufficiently small $\|Z\|_{C^1}$, both Γ_i remain hyperbolic under $(L_1)_* Y_i$.*

4.2.2. *L_2: global coherence.*

Definition 4.14 (L_2). The *coherence defect* $\mathcal{E}(x) = d_{S^{(2)}}(\Psi(\phi^{\Delta t}(x)), \psi^{(2),\Delta t/\lambda}(\Psi(x)))^2$ measures failure of semi-conjugacy. Set $\mathcal{V}(x) = V(x) + \alpha_0\, \mathcal{E}(x)$, $\alpha_0 > 0$, and $L_2 = \exp(-\nabla\mathcal{V})$.

Proposition 4.15 (L_2 fixed points). *$L_2(x) = x \iff x \in \Gamma$. Moreover $\mathcal{E}|_\Gamma = 0$ and $\nabla\mathcal{V}|_\Gamma = 0$.*

Proof. $\nabla\mathcal{V} = 0$ iff $\nabla V = 0$ (i.e. $x \in \Gamma$) and $\nabla\mathcal{E} = 0$. On Γ the semi-conjugacy holds (Proposition 4.6) so $\mathcal{E}|_\Gamma = 0$ and $\nabla\mathcal{E}|_\Gamma = 0$. □

4.2.3. L_3*: anti-collapse.*

Definition 4.16 (L_3)**.** For $\varepsilon_0 > 0$, define the regularized potential $U_{12}^{\varepsilon}(x) = ((d_g(x,\Gamma_1)^2 + \varepsilon_0)^{-1} + (d_g(x,\Gamma_2)^2 + \varepsilon_0)^{-1})$, cutoff χ_{12} in the inter-orbit region, $Q(x) = \chi_{12}(x)\nabla U_{12}^{\varepsilon}(x)$, and $L_3 = \exp(\eta Q)$ for small $\eta > 0$.

Proposition 4.17 (L_3 non-collapse and hyperbolicity)**.** *For sufficiently small* η*:* $d_g(L_3(\Gamma_1), L_3(\Gamma_2)) \geq d_g(\Gamma_1,\Gamma_2)$*, and each* Γ_i *remains a hyperbolic limit cycle of* $(L_3)_*Y_i$*.*

Proof. Q points in the direction increasing $d_g(\Gamma_1,\Gamma_2)$ to first order. Hyperbolicity is an open C^1 condition; L_3 is C^1-close to the identity for small η. □

4.3. R**-operators: resonance.**

4.3.1. R_1*: detection.*

Definition 4.18 (R_1)**.** Systems $\mathfrak{D}_1, \mathfrak{D}_2$ are in *$k{:}m$ period resonance* if $kT_1^* = mT_2^*$; in *Lyapunov resonance* if $\mu_{\max,1} = \mu_{\max,2}$; in *threshold resonance* if $\tau_1 = \tau_2$. Define

$$R_1(\mathfrak{D}_1,\mathfrak{D}_2) \coloneqq \{(k,m) : kT_1^* = mT_2^*\} \cup \mathbf{1}[\mu_{\max,1} = \mu_{\max,2}] \cup \mathbf{1}[\tau_1 = \tau_2].$$

$R_1 = \emptyset$ iff no resonance; only trivial coproduct unification is then possible.

Proposition 4.19 (Genericity)**.** *Each resonance locus is codimension-*1 *in* $\mathcal{P}^2$ *and has measure zero. Within each Arnold tongue, $k{:}m$ resonance is structurally stable under coupling.*

4.3.2. R_2*: amplification.*

Definition 4.20 (R_2)**.** Phase mismatch: $W_{k:m}(\theta_1,\theta_2) = (k\varphi_1(\theta_1) - m\varphi_2(\theta_2))^2$. Define $Z_{k:m} = -\nabla_{(x_1,x_2)} W_{k:m}$ and $R_2 = \exp(Z_{k:m})$.

Proposition 4.21 (Phase locking and descent)**.** $\mathrm{Fix}(R_2) \cap (\Gamma_1 \times \Gamma_2) = \{k\varphi_1 = m\varphi_2\}$*, consisting of* $\mathrm{lcm}(k,m)$ *points generically. Along the flow:* $\frac{d}{dt}W_{k:m} = -\|\nabla W_{k:m}\|^2 \leq 0$.

Proof. Fixed points are zeros of $Z_{k:m}$, i.e. critical points of $W_{k:m}$. On the torus $(\mathbb{R}/\mathbb{Z})^2$, the equation $k\varphi_1 - m\varphi_2 = 0$ has $\mathrm{lcm}(k,m)$ solutions generically. Descent: $\dot{W}_{k:m} = -\|\nabla W_{k:m}\|^2$ by definition. □

4.3.3. R_3*: stabilization.*

Definition 4.22 (R_3)**.** Resonance manifold: $\mathcal{M}_{k:m} = \{(x_1,x_2) \in \mathcal{N}(\Gamma_1) \times \mathcal{N}(\Gamma_2) : k\varphi_1(x_1) = m\varphi_2(x_2)\}$. Define $R_3 = \exp(-\nabla d_g(\cdot,\mathcal{M}_{k:m})^2)$.

Proposition 4.23 (Exponential stability)**.** *With* $U = d(\cdot,\mathcal{M}_{k:m})^2$*:* $\dot{U} = -4U$ *along* R_3*-flow, so* $U(t) = U(0)e^{-4t}$*. Both* Γ_i *are fixed by* R_3*.*

Proof. $\dot{U} = -\|\nabla U\|^2 = -4d^2 = 4U$ since $\|\nabla d\| = 1$ away from the cut locus. Orbit invariance: $\nabla d(\cdot,\mathcal{M}_{k:m})$ is tangent to each Γ_i on Γ_i, so the gradient flow moves Γ_i along itself. □

4.4. U**-operators: unification.**

4.4.1. *Admissible spans and pushout existence.*

Definition 4.24 (Resonant orbit and admissible span)**.** $\Gamma_\lambda \coloneqq \mathcal{M}_{k:m} \cap (\Gamma_1 \times \Gamma_2)$ (consists of $\operatorname{lcm}(k,m)$ points). A span $\mathfrak{D}_1 \xleftarrow{p_1} \mathfrak{D}_\lambda \xrightarrow{p_2} \mathfrak{D}_2$ is *admissible* if: (1) p_1, p_2 are dm3 morphisms; (2) smooth embeddings; (3) $\mathfrak{D}_\lambda$ has tubular neighborhoods in each S_i; (4) $V_1 \circ p_1 = V_2 \circ p_2$ on $\mathfrak{D}_\lambda$.

Proposition 4.25 (Projections are morphisms)**.** *The projections $p_i \colon \mathcal{M}_{k:m} \to S_i$ satisfy (M1)–(M4) with $p_i(\Gamma_\lambda) = \Gamma_i$.*

Proof. (M1): $p_i(\Gamma_\lambda) = \Gamma_i$ by definition. (M2): $k{:}m$ resonance gives semi-conjugacy with $\lambda = k/m$. (M3): $V_i \circ p_i$ equivalent near Γ_λ by transversality of $\mathcal{M}_{k:m}$ to V_i-level sets. (M4): bounded noise coefficients. □

Proposition 4.26 (Pushout existence)**.** *If the span is admissible, the pushout $\mathfrak{D}_{12} = \mathfrak{D}_1 \sqcup_{\mathfrak{D}_\lambda} \mathfrak{D}_2$ exists in* **dm3***, unique up to dm3-isomorphism.*

Proof. Conditions (1)–(3) give smooth gluing of S_1,S_2 along $p_i(\mathcal{M}_{k:m})$ by the smooth gluing theorem [10]; (4) gives smooth V_{12}. Hyperbolicity of Γ_{12} follows from hyperbolicity of Γ_1, Γ_2 and transversality of the gluing; uniqueness is categorical. □

4.4.2. *U_1, U_2, U_3.*

Definition 4.27 (U_1, U_2, U_3)**.** $U_1(\mathfrak{D}_1, \mathfrak{D}_2) \coloneqq \mathfrak{D}_1 \sqcup_{\mathfrak{D}_\lambda} \mathfrak{D}_2$.

The *resonance correspondence* $_{12} = \{(x_1, x_2) \in S_1 \times S_2 : (x_1, x_2) \in \mathcal{M}_{k:m}\}$ defines $U_2(x_1) = \pi_2(\pi_1^{-1}(x_1) \cap_{12})$.

The *synthesized orbit* $\Gamma_{\mathrm{syn}} = \omega$-limit set of generic trajectories; $U_3 = \exp(-\nabla d_g(\cdot, \Gamma_{\mathrm{syn}})^2)$.

Proposition 4.28 (dm3 closure under U_3)**.** *After applying U_3, the system $\mathfrak{D}_{12} = (S_{12}, g_{12}, Y_{12}, \Gamma_{\mathrm{syn}}, V_{12}, \sigma_{12})$ satisfies Axioms 1–8.*

Proof. Γ_{syn} is a limit cycle by the $k{:}m$ resonance collapsing the product torus; hyperbolicity from normal hyperbolicity of $\mathcal{M}_{k:m}$ (Proposition 4.23). Axioms 1–4 hold by construction of V_{12}; Axiom 5 inherited with $\tau_{12} \le \min(\tau_i)$; Axioms 6–8 from the above. □

4.5. **Proof of Theorem B.**

Proof of Theorem 1.2. (i) Well-definedness: $R_1 \neq \emptyset$ (resonance assumed); R_2 drives to phase locking (Proposition 4.21); R_3 stabilizes $\mathcal{M}_{k:m}$ (Proposition 4.23); admissibility gives the pushout (Proposition 4.26); U_1–U_3 proceed as in Definition 4.27; $\mathfrak{D}_{12} \in$ **dm3** by Proposition 4.28. (ii)–(iii) Period: $T_{12}^* = kT_1^* / \gcd(k,m)$ (smallest $T > 0$ with $T/T_i^* \in \mathbb{Z}$). Stability: slowest contraction dominates. Threshold: $c_{12} = \min(c_i)$, $\kappa_{12} = \max(\kappa_i)$, giving $\tau_{12} = \sqrt{c_{12}/\kappa_{12}} \le \min(\tau_i)$. □

5. Boundary operators and constraint geometry

5.1. B_1: **identity boundary.**

Definition 5.1 (B_1)**.** $B_1 \coloneqq \mathcal{W}^s(\Lambda) = \partial\mathcal{B}(\Gamma)$, the stable manifold of the boundary invariant set Λ. The critical Lyapunov level $c^* \coloneqq \sup\{c : \{V < c\} \subset \mathcal{B}(\Gamma)\}$ gives inner approximation $\{V = c^*\} \subseteq B_1$.

Proposition 5.2 (B_1 invariance)**.** $\phi^t(B_1) = B_1$ *for all $t \in \mathbb{R}$.*

Proof. $B_1 = \mathcal{W}^s(\Lambda)$ is invariant by definition of stable manifolds. □

5.2. B_2**: transition boundary.**

Definition 5.3 (B_2 and Arnold tongues)**.** Parameter space: $\mathcal{P} = \mathbb{R}_{>0} \times \mathbb{R}_{<0} \times \mathbb{R}_{>0}$. Transition boundary in $\mathcal{P}^2$:

$$B_2 = \bigcup_{k,m} \{kT_1^* = mT_2^*\} \cup \{\mu_{\max,1} = \mu_{\max,2}\} \cup \{\tau_1 = \tau_2\}.$$

Arnold tongue: $\mathcal{A}_{k:m} = \{|kT_1^* - mT_2^*| < \varepsilon_{k:m}(A)\}$. Inside $\mathcal{A}_{k:m}$: R-operators valid. Outside: invalid.

5.3. B_3**: collapse boundary.**

Definition 5.4 (B_3)**.** $U_{12}^{\varepsilon}(x) = (d_g(x, \Gamma_1)^2 + \varepsilon_0)^{-1} + (d_g(x, \Gamma_2)^2 + \varepsilon_0)^{-1}$. Define $C_{12}^* = \sup\{c : \dot{U}_{12}^{\varepsilon} > 0 \text{ whenever } U_{12}^{\varepsilon} > c\}$ and $B_3 = \{U_{12}^{\varepsilon} = C_{12}^*\}$.

Proposition 5.5 (Smoothness of B_3)**.** *For generic ε_0, B_3 is a smooth codimension-1 submanifold.*

Proof. U_{12}^{ε} is smooth; Sard's theorem gives generic regular values; the regular level set theorem applies. ☐

5.4. **Admissible regions and operator grammar.**

Definition 5.6 (Admissible regions)**.** $\mathcal{D}_S := S \setminus (B_1 \cup B_3)$; $\mathcal{D}_{\mathcal{P}} := \bigcup_{k,m} \mathcal{A}_{k:m}$. A configuration $(\mathfrak{D}_1, \mathfrak{D}_2)$ in states (x_1, x_2) is *fully admissible* if $(x_i) \in \mathcal{D}_{S_i}$ and $(\iota(\mathfrak{D}_i)) \in \mathcal{D}_{\mathcal{P}}$.

Theorem 5.7 (Operator grammar)**.** *Let $(\mathfrak{D}_1, \mathfrak{D}_2)$ be fully admissible. The sequence $L_3 \to L_1 \to g_1 \to L_2 \to g_2 \to R_1 \to R_2 \to R_3 \to U_1 \to U_2 \to U_3$ is well-defined and produces a dm3 system at every stage. Ordering constraints: (1) L_3 before g_1 (separation before expansion); (2) L_1 before L_2 (local before global coherence); (3) g_1 before g_2 (base before recursive expansion); (4) R_1 before U_1 (resonance needed for pushout); (5) R_3 before U_3 (resonance stabilized before synthesis). Steps involving L_1 and g_2 are conditional on Conjectures 4.13 and 4.7. The reduced sequence $L_3 \to g_1 \to R_1 \to R_2 \to R_3 \to U_1 \to U_2 \to U_3$ holds unconditionally.*

Proof. dm3 membership at each stage: After L_3: Proposition 4.17. After L_1: Proposition 4.12 (conditional). After g_1: Proposition 4.3. After L_2: Proposition 4.15. After g_2: semi-conjugacy preserved (conditional on Conjecture 4.7). After R_1: detection only. After R_2, R_3: Propositions 4.21, 4.23. After U_1: Proposition 4.26. After U_2: correspondence, no dm3 alteration. After U_3: Proposition 4.28. Admissibility maintained since B_1, B_3 persist under controlled operator perturbations (shown case by case above). ☐

6. Contact geometric physics

6.1. **Why contact geometry.** Seven structural features of dm3 systems force the contact framework: (1) hyperbolic limit cycle attractors; (2) strictly decreasing Lyapunov functions; (3) gradient-like operators; (4) noise-as-fuel stochastic stability; (5) embodiment threshold as bifurcation parameter; (6) Arnold tongue resonance; (7) categorical pushouts of dissipative systems. Features (1)–(2) exclude Hamiltonian mechanics (Liouville's theorem). Features (4)–(5) require a dissipation parameter evolving along trajectories. Contact geometry is the unique framework satisfying all seven simultaneously.

6.2. **The contact manifold.**

Definition 6.1 (Extended state space and contact form)**.** Define $M = S \times \mathbb{R}$ with coordinates (s, z), where z is the *action variable*. Let $\lambda \in \Omega^1(S)$ with $d\lambda = \omega$ a closed non-degenerate 2-form. Set $\alpha = dz - \lambda$.

Proposition 6.2 (Contact structure and Reeb field)**.** *(M, α) is a contact manifold with $R_\alpha = \partial_z$.*

Proof. $\alpha \wedge (d\alpha)^n = (dz - \lambda) \wedge (-\omega)^n \neq 0$ since $\omega^n \neq 0$ and dz is independent of S-forms. For Reeb: $\alpha(\partial_z) = 1$ and $\iota_{\partial_z}(-\omega) = 0$. □

6.3. **Winding integral and Theorem A claim (iv).**

Proposition 6.3 (Topological invariance of $\oint_\Gamma \lambda$)**.**
(i) *$\oint_\Gamma \lambda$ is independent of the choice of λ with $d\lambda = \omega$.*
(ii) *It is invariant under admissible operators fixing Γ or mapping it to a homologous cycle.*
(iii) *It is nonzero when Γ bounds no chain in S.*

Proof. (i) $\oint_\Gamma df = 0$ for any f since Γ is closed. (ii) A contactomorphism changes λ by an exact form along Γ. (iii) Stokes' theorem. □

6.4. **Contact Hamiltonian and dynamics.**

Definition 6.4 (Generative Hamiltonian and equations)**.** $H(s, z) = H_{\text{phase}}(s) + H_{\text{diss}}(s, z)$, where $H_{\text{phase}} = \frac{1}{2}\omega(Y, Y)$ and $H_{\text{diss}} = -\gamma V(s) e^{-\beta z}$ for $\gamma, \beta > 0$. The full dynamics are $\dot{x} = X_H(x) + X_{\mathcal{R}}(x)$, where $\mathcal{R} = \frac{c}{2} \|\dot{s} - Y(s)\|_g^2$ is the Rayleigh function and $X_{\mathcal{R}}(s) = -c(\dot{s} - Y(s))$.

Proposition 6.5 (Recovery of dm3 dynamics)**.** *$\pi_* X_H(s) = Y(s) - \gamma e^{-\beta z} \nabla V(s) + O(V)$. On Γ: $\pi_* X_H = Y$. Along trajectories: $\dot{V} \leq -c \|\nabla V\|^2 \leq 0$, with equality iff $s \in \Gamma$.*

6.5. **Action functional and Euler–Lagrange equations.**

Definition 6.6 (Generative action)**.** $\mathcal{S}[\gamma] = \int_0^T (\dot{z} - \lambda(\dot{s}) - H(s, z))\, dt$ for $\gamma(t) = (s(t), z(t)) \colon [0, T] \to M$.

Theorem 6.7 (Contact Euler–Lagrange equations)**.** *Critical points of $\mathcal{S}$ satisfy*

$$\dot{s} = \frac{\partial H}{\partial p} + X_{\mathcal{R}}, \quad \dot{z} = \lambda(\dot{s}) + H, \quad \dot{p} = -\frac{\partial H}{\partial s} + p\frac{\partial H}{\partial z}.$$

The contact term $p\,\partial H/\partial z$ is absent from standard Hamiltonian mechanics.

Proposition 6.8 (On-orbit action)**.** *$\mathcal{S}[\gamma|_\Gamma] = \oint_\Gamma \lambda$ (the topological invariant of Proposition 6.3).*

6.6. **Contact Noether theorem and winding number conservation.**

Theorem 6.9 (Contact Noether theorem)**.** *If $\{\Phi_\varepsilon\}$ is a one-parameter family of contactomorphisms with generator W satisfying $\mathcal{L}_W \alpha = f\alpha$ and $\mathcal{L}_W H = fH$, then $J_W = \alpha(W)$ satisfies $\dot{J}_W = f \cdot J_W$ along X_H-trajectories. For strict contactomorphisms ($f = 0$): J_W is conserved.*

Corollary 6.10 (Winding number conservation)**.** *Phase rotation on Γ is a strict contactomorphism. Its conserved quantity is $\oint_\Gamma \lambda$—the topological invariant making Γ structurally stable.*

6.7. Boundaries as contact submanifolds.

Proposition 6.11 (B_1 and B_3 in contact geometry). *$\hat{B}_1 = B_1 \times \mathbb{R} \subset M$ is a contact hypersurface ($\alpha|_{\hat{B}_1}$ is a contact form). $\hat{B}_3 \subset M$ is a Legendrian submanifold ($\alpha|_{\hat{B}_3} = 0$, since $H|_{B_3} = 0$).*

6.8. Proof of Theorem C.

Proof of Theorem 1.3. Step 1 (Floquet coordinates). By Floquet theory [6], there exist coordinates (ρ, θ) near Γ in S with $\dot{\rho} = \mu_{\max}\rho + O(\rho^2)$ and $\dot{\theta} = \omega + O(\rho)$.

Step 2 (Contact extension). Extend to $M = S \times \mathbb{R}$ with H from Definition 6.4. The contact equations via Theorem 6.7 yield: $\dot{\rho} = \mu_{\max}(1 - e^{-\beta z})\rho + O(\rho^2)$, $\dot{\theta} = \omega + O(\rho)$, $\dot{z} = \omega - |\mu_{\max}|\,\rho^2 e^{-\beta z} + O(\rho^3)$.

Step 3 (Contactomorphism). The coordinate change from the original dm3 coordinates is a contactomorphism: it preserves α up to a nonvanishing factor e^f, preserving all contact structure.

Step 4 (Uniqueness). The normal form depends only on $(\mu_{\max}, \omega, \beta) = \iota(\mathfrak{D})$. Any two dm3 systems with the same canonical triple are locally contact-diffeomorphic. □

7. Structural stability

7.1. Single-operator and boundary persistence.

Proposition 7.1 (Persistence under operators). *For sufficiently small operator parameters: $g_i(\mathfrak{D}) \in$ **dm3**; $L_i(\mathfrak{D}) \in$ **dm3**; $R_i(\mathfrak{D}_1, \mathfrak{D}_2) \in$ **dm3** $\times$ **dm3**; $U_i(\mathfrak{D}_1, \mathfrak{D}_2) \in$ **dm3**.*

Proof. Each operator is C^1-close to the identity or a dm3 morphism near Γ; hyperbolicity (open C^1 condition) and Lyapunov descent (continuous in C^1) are preserved. See Propositions 4.3, 4.17, 4.23, 4.28. □

Proposition 7.2 (Boundary persistence). *Under sufficiently small C^1 perturbations: B_1', B_2', B_3' exist and vary C^1-smoothly.*

Proof. B_1: stable manifold theorem applied to perturbed Λ'. B_2: implicit function theorem on $kT_1^* - mT_2^* = 0$. B_3: regular level set theorem on U_{12}^{ε}. □

7.2. Proof of Theorem D.

Proof of Theorem 1.4. Step 1. Each operator produces a C^1-perturbation of Y of size δ_i (bounded by its parameter; see Definition 4.1 for g_1: $O(\delta)$; L_1, L_3: $O(\varepsilon)$, $O(\eta)$; R_2, R_3: $O(\|W_{k:m}\|_{C^1})$).

Step 2. The composed perturbation satisfies $\|Y_N - Y\|_{C^1} \leq \prod_{i=1}^{N}(1 + \delta_i) - 1$. Choosing parameters so this product satisfies $\prod(1 + \delta_i) - 1 < \varepsilon_0$ keeps the composition within the ε_0-ball.

Step 3. Within the ε_0-ball, Shub's theorem [13] gives a unique hyperbolic Γ' near Γ with $d_{C^1}(\Gamma', \Gamma) = O(\varepsilon_0)$. The winding number is preserved by Proposition 6.3.

Step 4. $T^{*\prime}$, $\mu_{\max}'$, τ' vary by $O(\varepsilon_0)$ (period: implicit function theorem; Floquet: spectral perturbation; threshold: continuity of $\sqrt{c/\kappa}$).

Step 5. Proposition 7.2.

Step 6. $\varepsilon_0 < \varepsilon_0$ is satisfied throughout, so Axioms 1–8 hold for $\mathfrak{D}'$ by Steps 3–5. □

Remark 7.3 (Stability radius). $\varepsilon_0 = \frac{1}{2}\,|\mu_{\max}|\cdot\sup_\Gamma \|\mathrm{Hess}\,V\|^{-1}$: larger $|\mu_{\max}|$ (faster contraction) and flatter V both enlarge the stability radius. This bound is computable from $\iota(\mathfrak{D})$ and the Lyapunov geometry.

8. Discussion

8.1. **Summary of main results.** This paper develops a geometric framework for dissipative dynamical systems with structured limit cycles. The dm3 system, formalized through eight axioms, is the central object. An operator algebra acting on the category **dm3** governs expansion (g-operators), coherence (L-operators), resonance (R-operators), and unification (U-operators). The entire structure is embedded in contact geometry on $M = S \times \mathbb{R}$.

The four main theorems establish: (A) existence and exponential stability of the dm3 orbit, with topological invariant $\oint_\Gamma \lambda$ and stochastic stability below τ; (B) closure of **dm3** under $\mathcal{U}$, with the prediction $\tau_{12} \le \min(\tau_i)$; (C) the universal contact normal form with parameters $(\mu_{\max}, \omega, \beta)$; and (D) C^1-structural stability with explicit radius ε_0.

8.2. **Relation to nonequilibrium thermodynamics.** The action variable z plays the role of entropy: it increases monotonically ($\dot z > 0$ on Γ) and the contact form $\alpha = dz - \lambda$ encodes the first and second laws simultaneously. The embodiment threshold τ functions as a noise-induced phase transition. The result $\tau_{12} \le \min(\tau_i)$ says that higher-order organization requires lower effective temperature, consistent with observations in biological and physical oscillator networks and with stochastic thermodynamic theory [7, 12].

8.3. **Relation to coupled oscillator theory.** Period resonance, Arnold tongues, and mode locking are classical [12, 9]. The contribution here is embedding these phenomena in a categorical structure: resonance is the precondition for the pushout in **dm3**. The prediction $\tau_{12} \le \min(\tau_i)$ is new and testable.

8.4. **Open problems.**

(1) **Global basin expansion** (Conjecture 4.4): conditions on S for global (not just local) basin expansion.
(2) **Invariant torus stability** (Conjecture 4.7): KAM theory (quasi-periodic) and Poincaré–Melnikov (resonant) in the abstract setting. Proved in [14] for the toy model.
(3) **Higher-order resonance and Devil's staircase**: measure-theoretic and fractal properties of the full Arnold tongue diagram in $\mathcal{D}_{\mathcal{P}}$.
(4) **Infinite-dimensional extensions**: Hilbert/Banach manifold generalizations; infinite-dimensional Floquet theory; PDE and field-theory dm3 systems.
(5) **Categorical completeness of dm3**: does **dm3** have all finite limits? colimits? Is it a topos?

8.5. **Companion paper.** The companion paper [14] instantiates every definition and operator on the explicit system $\dot r = r(1 - r^2) + 2(r-1)e^{-z}$, $\dot\theta = 1$, $\dot z = r^2 - 2(r-1)^2 e^{-z}$ on $S = \mathbb{R}^2_{>0}$, verifying all theoretical predictions including the global attractor, four bifurcations, stationary SDE measure, and Conjecture 4.7 in the 1:2 resonant case.

Appendix A. The category **dm3**: formal development

[Full categorical development: objects, morphisms, composition law, associativity, identity, pushouts (proved in Proposition 4.26), coproducts (disjoint union when $R_1 = \emptyset$), and initial/terminal objects if they exist. Approximately 3–4 pages.]

Appendix B. Supporting lemmas

[Proofs deferred from the main text. Approximately 2–3 pages.]

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